# MesoNet: A Spatio-Temporal Neural Architecture for Massively Energy-Efficient Unsupervised Learning

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(Project Headboard)

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# VISUAL SUMMARY

**SNNs face issues with information retention and deeply scaling**

**Developed a model to uncover deep SNN learning dynamics**

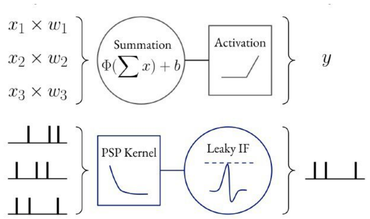
**Variable Plasticity and Triangulated Attribution to remedy issues in network mode**

**MesoNet lies within 5.05% points of state-of-the-art ANNs while consuming half (47.23%) the energy**

# BACKGROUND

Artificial Neural Network (ANN) **clustering algorithms** require hundreds of watts, considerably limiting **energy critical applications**. A promising solution, SNNs require mere *milliwatts* on **neuromorphic chips.**

## Spiking Neural Networks (SNNs)



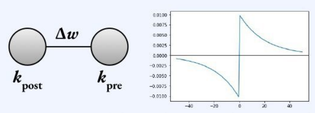

**Spiking Neural Networks (SNNs)** closely mimic biological neurons by utilizing discrete events called "spikes" to represent information.

Figure 1. A comparison of ANNs (top) vs. SNNs (bottom) [Credit: *Krishna Bhatt*]

## Definition: STDP Learning Rule

Given a synaptic weight $w \in \mathbb{R}$ and spike timing difference $\Delta t = t_{post} - t_{pre}$, the weight update is given as:

$$\Delta w = \begin{cases} A_+ e^{-\frac{\Delta t}{\tau_m}}, & \text{if } \Delta t > 0 \\ A_- e^{\frac{\Delta t}{\tau_m}}, & \text{if } \Delta t < 0 \end{cases}$$




where $A_+$, $A_-$, and $\tau_m$ are learning parameters. This rule strengthens causal relationships and weakens anti-causal ones, enabling bio-plausible local learning.

Figure 2-3. Poisson-based STDP Algorithm Features [Credit: *Krishna Bhatt*]

## Research Gap and Approach

Little work has been done to uncover **internal learning behavior**. STDP, the prevalent learning rule, struggles to scale from toy datasets to practical tasks, **reducing accuracy by up to 40%** compared to ANNs (Deng, et al.)

**1907** Louis Lapicque describes the first LIF neuron model

**1952** Hodgkin and Huxley discover the voltage dynamics of a squid axon

**1993** Henry Markram invents STDP based on Hebbian learning

**2017** Kheradpishesh develops the convolution SNN

**2021** Zhang develops the bifurcation neuron model

### Challenges

- SNNs face 'catastrophic forgetting' with large inputs.
- STDP rarely enables deeper layers to generalize on lower level features.
- No viable real-world use case beyond toy datasets (MNIST).

### My Approach

- Uncovered nonlinear learning dynamics in SNN systems.
- Investigated information loss and chaos in deep layers.
- Developed a novel **plasticity based** learning algorithm and mathematically proved its improved capability

# PROBLEM FRAMEWORK

Deconstructing deep multilayer SNN learning requires evaluation with dynamical systems and chaos theory, which explain how systems evolve.

**The Leaky Integrate and Fire Neuron Model**

$$\tau_m \frac{dV}{dt} = -(V(t) - V_{\text{rest}}) + R(I(t))$$

**Definition: The Multi-layered LIF-SNN**

Consider an $L$-layer feed-forward spike neural network with width $N$ with synaptic weight matrices $\mathbf{W}^l \in \mathbb{R}^{N_x N}$, membrane constants $\boldsymbol{\tau}_m^l$, post-synaptic potential $\epsilon(t)$, and input synaptic current $\boldsymbol{I}(t) = (I_1(t), \ldots, I_M(t))^\top$ denotes the $M$-dimensional input signals.

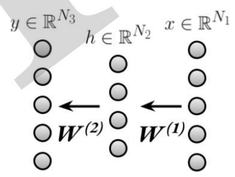



Figure 4. The basic 3-layered network for preliminary study [Credit: *Krishna Bhatt*]

# EXPLORING INTERNAL LEARNING DYNAMICS

**STDP learning forms a feedback loop** (Δw→Δt→Δw). To model the base case deterministically, I developed a novel spike approximation technique.

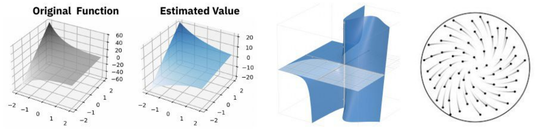



Figure 5-7. Q-Spike approx., $w^*$ distribution [*Krishna Bhatt*], Dissipative system [*wikipedia.com*]

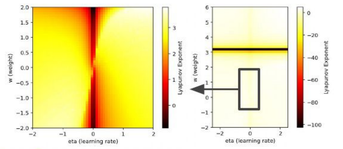



Figure 8. Lyapunov exp. heatmap [Credit: *Krishna Bhatt*]

Lifting from the basic (1:1) to multilayer case, mapping Lyapunov exponents revealed strange attractors as well as **stability at the locus** of parameter options.

## Guiding Principle

The bifurcation parameter $\eta$ defines whether the system converges, **learns efficiently at the edge of chaos**, or diverges. Optimal learning occurs in the narrow regime where **Lyapunov exponents are close to 0**.

## Supporting Theorems

### Theorem 1: Quadratic Spike Approximation.

For any given LIF neuron with a membrane voltage $V$ and postsynaptic potential $\epsilon$, if given an input parameter $\lambda$, the membrane voltage can be generalized as:

$$f(x) = w_j I e^{-\frac{x}{\tau_m}}$$

where the spike timing difference is given by $\Delta t(w)$ with a Taylor expansions truncated at the $N$-th ordered term such that the $N$ satisfies:

$$t' - t = x = -\tau_m \omega(-\frac{\theta}{w_j k \tau_m}) \qquad \min_{N \in \mathbb{Z}} (N+1)! \geq 8e\tau_m^2$$

### Theorem 2: Bifurcation Classification.

A multilayer STDP-LIF network is an entirely dissipative system composed of unstable critical points $w^*$. Solving for local linearity in the neighborhood of points $w^*$,

$$w^* = (\theta \cdot \exp(\frac{k}{\tau_m})/k^2 \qquad f'(w^*) = \eta(\frac{n}{\tau_m w^* k(1 - \frac{k}{\tau_m})})$$

where in the triangular-block Jacobian, each represents the eigenvalues, thus making the system contain a stability-switch bifurcation (for $\eta$) within codimension-1. A candidate Lyapunov energy function modeling kinetic and potential interactions is provided as:

$$E(w, \eta) = \frac{1}{2}[\Delta t(w)]^2 + \frac{\lambda}{2}(f'(w))^2$$

# ENHANCING SNN LEARNING

## Breakthrough 1: Variable Plasticity

To ensure that the network **selectively retains important information**, I designed a bio-plausible plasticity mechanism where learning rate decays naturally over time, increasing with frequent weight updates.

$$\tau_d \frac{d\eta}{dt} = -(\eta - \eta_{\min}) + \frac{1}{(\eta_{\max} - w(\Delta t))}$$

Coupled with **synaptic flipping** mechanism where synapses can switch between excitatory and inhibitory at $\eta = 0$ and $w = 0$.

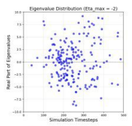


Figure 9. $\eta_{max}$ and $\eta_{min}$ can now directly control network e-vals. (Randomly sampled eigenvalues of a 3 neuron network) [Credit: *Krishna Bhatt*]

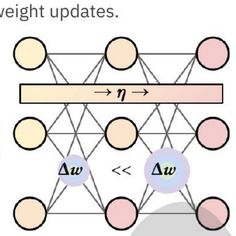
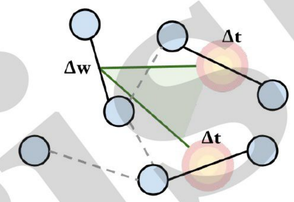



Figure 10. Plasticity is modulated by frequency of weight updates. [Credit: *Krishna Bhatt*]

## Breakthrough 2: Triangulated Attribution

Variable plasticity along with attractor/repeller dynamics leads to **runaway behavior and slow convergence**. I created an external weight attribution algorithm for the network to **accurately propagate a global error signal.**



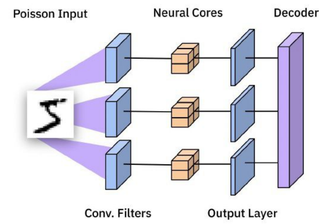

$$R_p^{i,j} = \eta_p \exp(-\frac{|w_{pi} - w_{pj}|}{\kappa})$$

Attribution depends on a **relevance score**, derived from the synapse's influence on external spiking activity.

Figure 11. Synapses triangulate a target synapse with high relevance. [Credit: *Krishna Bhatt*]

## Implementation

**Split-&-Merge modules** divide input images into multiple pathways for processing by different convolutional filters. Within them, I designed **neural cores to develop diverse representations**.




(1) Improved multi-scale feature extraction
(2) Enhances network parallelization

Figure 12. A simplified diagram of the MesoNet architecture (Poisson Input → Convolutional Filters → Neural Cores → Output Layer → Decoder) [Credit: *Krishna Bhatt*]

Triangulation was optimized through use of **Q-spike approximation** (Theorem 1) and **Dijkstra's Algorithm** to find nodes.

Simulated with... BRIAN NVIDIA CUDA

[Credit: *Respective Logos*]

## MesoNet: Central Theorem

Considering a fully connected multilayer LIF-STDP network incorporating variable plasticity and triangulated attribution, the fixed points and the Jacobian of the locally linearized dynamics, respectively would be given by

$$\eta^* = \eta_{\min} + \frac{1}{\eta_{\max} - w^*} \qquad J_{\text{local}} = \begin{pmatrix} -\frac{\eta^*}{\tau_m} A + \epsilon g_w & 0 \\ -\frac{1}{\tau_d(\eta_{\max} - w^*)^2} & -\frac{1}{\tau_d} \end{pmatrix}$$

For large networks, the Jacobian spectrum follows a free convolution of semicirular laws centered at $\lambda_1$, given below, with a stable eigenvalue at $\lambda_2$, thus allowing the network to freely alter modes from a saddle-type bifurcation and a stable manifold.

$$\lambda_1 = -\frac{\eta^*}{\tau_m} \cdot \frac{\omega(-\frac{\theta}{wk\tau_m})}{\tau_m w^2 k(1 + \omega(-\frac{\theta}{wk\tau_m}))} + \epsilon g_w$$

The instability is dominated by $1/w^2$ which shrinks as $w$ approaches $w^*$, developing stability.

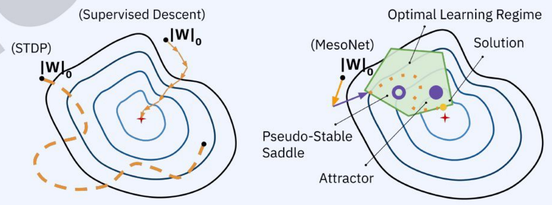



Figure 13. Saddle points and attractors are dynamically created and annihilated to "lure" weights in the optimal direction without awareness of the gradient. [Credit: *Krishna Bhatt*]

## Emergent Properties

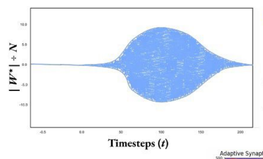



Figure 14. Constrained bifurcation diagram of a 3 neuron network with MesoNet learning algorithm.

**Bifurcation Plot:** The repulsive force from triangulated attribution surpasses instability, triggering onset of a controlled convergence.

**Turing Patterns:** A smoothed Turing map (500x500) revealed that MesoNet forms cortical-like organization, unlike standard STDP.

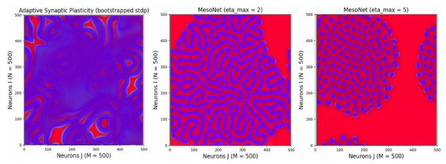



Figure 15. Turing map of inhibitory (red) /excitatory (blue) interactions (STDP on the left, MesoNet being the right two) [Credit: *Krishna Bhatt*]

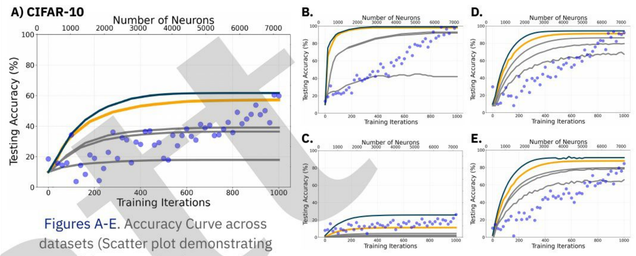
## Iterative Development

| Attempted to use SVD to Decouple Spike Timing Relationships | Tried decomposing global signals with Fourier Transform for attribution | Overlooked certain redundancies in regards to plasticity updates. | Faced compute limits while running heavy workloads on personal PC. |
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# PERFORMANCE VALIDATION

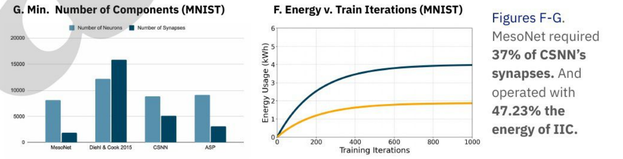
Rigorously tested success criteria: **accuracy, robustness, efficiency**

- **Datasets:** A. CIFAR-10, B. MNIST, C. CIFAR-100, D. F-MNIST, E-MNIST
- **SOTA Contenders:** Convolution SNN (SLAYER), Adaptive Synaptic Plasticity, Simple, **Invariant Information Clustering (IIC)**, **MesoNet**




Figures A-E. Accuracy Curve across datasets (Scatter plot demonstrating minimum required neurons)

MesoNet surpasses the current SOTA SNN, achieving a **56.6% accuracy on CIFAR-10**, 18.57% points above the previous model.




Figures F-G. MesoNet required **37% of CSNN's synapses.** And operated with **47.23% the energy of IIC.**

Phenomenal resource efficiency, demonstrating the enhanced synaptic ability to generalize high-level information. [All graphs generated by *Krishna Bhatt*]

# APPLICATIONS & IMPACT



[Credit: *wikipedia.org*]

**Medical Implants.** MesoNet can be utilized in implants like pacemakers to assist patients for *over 15 years*.



Credit: [*nasa.gov*]

**Deep Space Missions.** MesoNet can reduce energy consumption on the Parker Solar Probe by *around 23%*, massively extending its operating lifespan.



Credit: [*thenextsummit.com*]

**Remote Sensing.** MesoNet can enhance the energy efficiency of battery-powered wildfire sensing cameras by nearly *40%*.

## Key Findings and Conclusions

### Enhanced multi-layer feature extraction

- Significant improvement in SNN for larger tasks (i.e. CIFAR-10)
- Comparable accuracy to SOTA ANNs (ie. ICC)

**5.1%** within range of the SOTA ANN model

**53%** less energy than the SOTA ANN (IIC)

### Novel Dynamical Systems Model of Learning

- Defined the full bifurcation dynamics underlying SNN learning processes.
- Unveiled the dynamics of signal propagation and supported the existence of an **optimal learning regime** defined by lyapunov exponents..

## Future Work

- Stricter restrictions regarding the the boundaries of the optimal regime. (Perhaps requiring dynamical isometry in the initializations?)
- Exploration into energy-efficient or favorable network geometries that emerge from variable plasticity.

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**NOTE:** All images created by researcher unless otherwise noted. Full theorems in notebook.

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